Om Satpute

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Education

Pune Institute of Computer Technology (PICT), Pune 2023 – Present
Bachelor of Engineering in Computer Science Current CGPA: **9.465/10**

Pride Academy and Junior College, Shrirampur 2022 – 2023
HSC (Class XII) Score: 80.33%

Experience

Research Intern Jan 2026 – Apr 2026
Pune Institute of Computer Technology (PICT) Pune, India

- Designed a CSP-based timetable engine in Node.js/Express.js with custom heuristics, generating conflict-free schedules for 500+ sessions.
- Built the MySQL/Sequelize backend with 10 relational models, 28 REST APIs, indexing, and transactions.
- Created React.js frontend with timetable views for divisions, teachers, halls, and labs with PDF export.
- **Tech Stack:** Node.js, Express.js, MySQL, Sequelize, React.js.

Projects

NovaDB – *Relational Database Management System*

- Built a **relational database management system** from scratch with a **custom SQL parser and CLI**.
- Implemented **SQL operations** including `CREATE`, `INSERT`, `SELECT`, `UPDATE`, `DELETE`, `DROP`, and `JOIN`.
- Developed **B+ Tree indexing** for faster query execution and efficient record retrieval.
- Added **PRIMARY KEY, FOREIGN KEY, UNIQUE, and NOT NULL constraints** to ensure data integrity.
- Implemented **robust error handling** for invalid queries, constraint violations, and runtime errors.
- **Tech Stack:** Java, File System, B+ Tree.

PetConnect – *Full-Stack MERN Platform*

- Built a **multi-role pet platform** for Users, Caretakers, and Admins with **JWT/Google OAuth, RBAC, and OTP-based password recovery**.
- Engineered **pet marketplace, caretaker bookings, admin moderation, and Cloudinary image management**, with **Razorpay** payments for transactions and premium subscriptions.
- Developed **real-time chat and notifications** using Socket.io and integrated **Google Gemini AI** with custom **rate limiting** for usage quotas.
- Automated **scheduled emails and background tasks** using Nodemailer and Cron jobs.
- **Tech Stack:** React.js, Vite, Tailwind CSS, Node.js, Express.js, MongoDB, Socket.io, Razorpay, Gemini AI, Cloudinary, Nodemailer, Cron Jobs.

EDUFLEX – *React.js, Node.js, Express.js, MongoDB, Socket.IO*

- Built a **Classroom Management Platform** with secure **Role-Based Access Control (RBAC)** for Admins, Teachers, and Students.
- Implemented secure **JWT Authentication Authorization** along with **Google OAuth** for user auth.
- Integrated **Socket.IO** for **real-time classroom communication and live quizzes**.
- Developed core features including **classroom management, attendance tracking, tests, and email services** using **Nodemailer**.

Technical Skills

Languages: C++, Java, JavaScript, SQL

Frontend: React.js, Redux Toolkit, Vite

Backend: Node.js, Express.js, REST APIs, JWT Authentication, Socket.io, WebRTC

Database: MongoDB, MySQL

Core Subjects: Data Structures & Algorithms, Core Java, OOP, DBMS, OS, CNS .

Problem Solving

Solved 600+ Data Structures and Algorithms problems across coding platforms.

Extracurricular Activities

- PICT IEEE Student Branch Member.
- Volunteered for Credenz Group Day
- Winner – College Cricket Championship, Season 3.